HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2013)

Student No.: 06 Name: _____ Marks: / 28

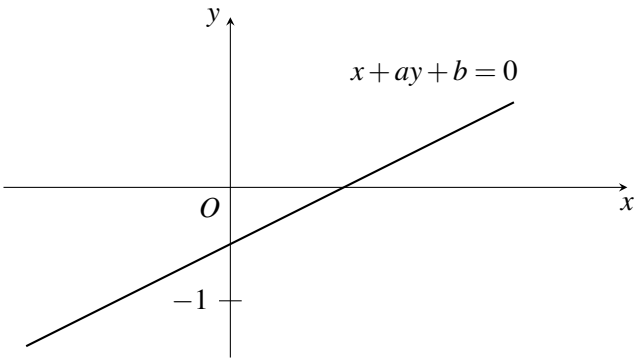
Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The figure shows $x+ay+b=0$. Which statements are true?

I. $a<0$ II. $b<0$ III. $a<b$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

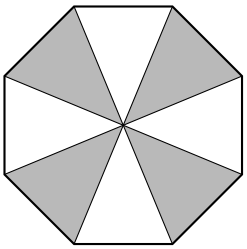
(DSE-2013 PII Q14)

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2. In the figure, the regular octagon is divided into eight identical isosceles triangles and four of them are shaded. The number of axes of reflectional symmetry of the octagon is



A. 2

B. 4

C. 8

D. 16

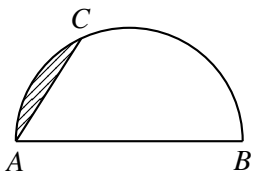
(DSE-2013 PII Q15)

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3. In the figure, the diameter of the semicircle ABC is 3 cm. If chord $AC = 2$ cm, find the area of the shaded region corrected to the nearest $0.01\,\text{cm}^2$.



A. $0.23\,\text{cm}^2$

B. $0.52\,\text{cm}^2$

C. $0.64\,\text{cm}^2$

D. $1.07\,\text{cm}^2$

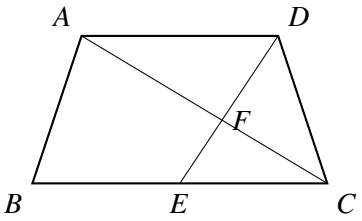
(DSE-2013 PII Q16)

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4. In the figure, $ABCD$ is a trapezium with $AD//BC$, and $AD : BC = 2 : 3$. Let E be the midpoint of BC. AC and DE intersect at F. If the area of ΔCEF is $36\,\text{cm}^2$, then the area of the trapezium $ABCD$ is.



A. $216\,\text{cm}^2$

B. $264\,\text{cm}^2$

C. $280\,\text{cm}^2$

D. $320\,\text{cm}^2$

(DSE-2013 PII Q18)

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5. If $0^\circ < x < 90^\circ$, which must be true?

I. $\tan x \tan(90^\circ - x) = 1$

II. $\sin x - \sin(90^\circ - x) < 0$

III. $\cos x + \cos(90^\circ - x) > 0$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2013 PII Q23)

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6. If the mean and mode of nine numbers $14, 6, 4, 5, 7, 5, x, y, z$ are 8 and 14 respectively, then the median of these nine number is

A. 5.

B. 6.

C. 7.

D. 8.

(DSE-2013 PII Q27)

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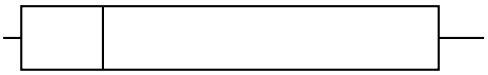
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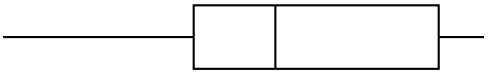
7. The stem-and-leaf diagram shows hourly wages. Which box-and-whisker diagram may represent them?

4	0 2 2 2 4 4 4 7
5	0 0 1 2 2 6 8 9
6	3 5 5 7
7	0
8	2 6
9	5

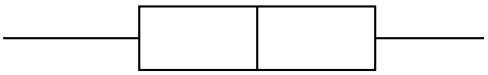
A.



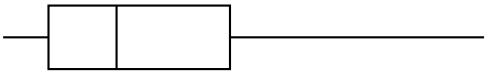
B.



C.



D.



(DSE-2013 PII Q29)

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8. $\text{A00000E00001}_{16} =$

A. $10 \times 16^{11} + 14 \times 16^5 + 17$

B. $11 \times 16^{11} + 15 \times 16^5 + 17$

C. $10 \times 16^{12} + 14 \times 16^6 + 272$

D. $11 \times 16^{12} + 15 \times 16^6 + 272$

(DSE-2013 PII Q33)

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9. If $x - \log y = x^2 - \log y^2 - 10 = 2$, then $y =$

A. 100

B. 2 or -4

C. $\frac{1}{100}$ or 10000

D. $\frac{1}{10000}$ or 100

(DSE-2013 PII Q34)

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10. The nth term of a sequence is $2n - 19$. Which are true?

I. 25 is a term of the sequence..

II. The sequence has 10 negative terms.

III. The sum of the first n terms of the sequence is $n^2 - 18n$.

A. I only

B. II only

C. I and III only

D. II and III only

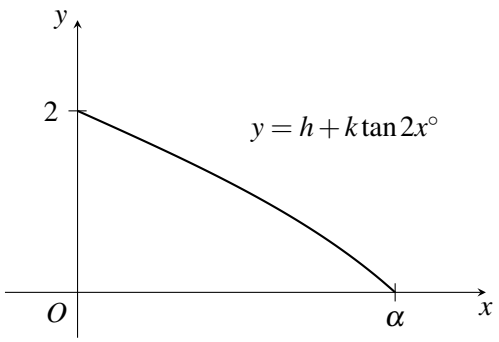
(DSE-2013 PII Q38)

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11. Let h and k be constants. The figure shows the graph of $y = h + k\tan 2x^\circ$, where $0 \le x \le \alpha$. Which of the following are true?



I. $h > 0$

II. $k < 0$

III. $\tan\alpha = \frac{1}{k}$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

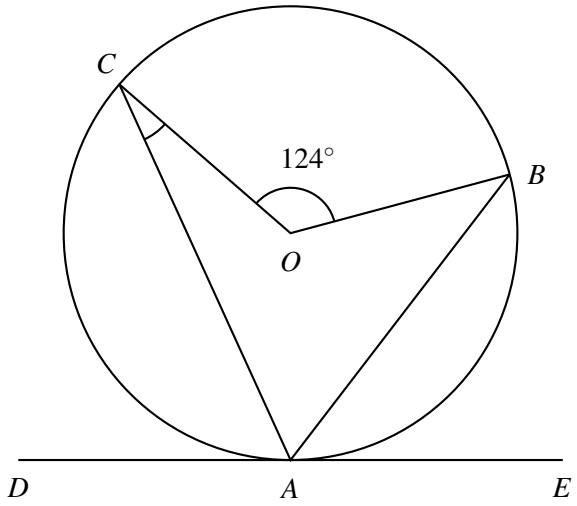
(DSE-2013 PII Q39)

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12. In the figure, O is the centre of the circle ABC, and DE is tangent to the circle at A. If AB is the angle bisector of $\angle CAE$, then $\angle ACO =$



A. 26°

B. 28°

C. 31°

D. 34°

(DSE-2013 PII Q41)

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13. Let O be the origin. If the coordinates of the points A and B are $(0, 12)$ and$(30, 12)$ respectively, then the $y-$coordinate of the circumcentre of ΔOAB is

A. 6.

B. 8.

C. 12.

D. 15.

(DSE-2013 PII Q43)

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14. If the variance of the five numbers x_1, x_2, x_3, x_4 and x_5 is 13, then the variance of the five numbers$3x_1 + 4, 3x_2 + 4, 3x_3 + 4, 3x_4 + 4$ and $3x_5 + 4$ is

A. 39

B. 43

C. 117

D. 121

(DSE-2013 PII Q45)

I can finish _____ questions

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